

A spatial analysis of SST & Chlorophyll Concentration of North Sea

Abstract

Sea Surface Temperature (SST) is the water temperature close to the ocean surface. It mainly varies from latitude, the warmest regions are those with low latitude. Chlorophyll concentration is an indicator for the amount of photosynthetic plankton or phytoplankton present in the ocean. Sea surface temperature & chlorophyll concentration is an important phenomenon to learn about ocean characteristics. Sea surface temperature & chlorophyll concentration may vary daily, monthly & seasonally. This study aims to find out the variation of sea surface temperature & chlorophyll concentration with using geographic information system of North sea. We use satellite data from many organizations & use Arc GIS software for our project data analysis. The results found the highest sea surface temperature was in the German region and the lowest sea surface temperature found in the UK region. The highest average chlorophyll concentration in the English channel, German & Danish part & lowest average chlorophyll concentration are found in the UK & Norwegian part. Sea surface temperature and the spread of chlorophyll through remote sensing can be used as determination of potential fishing zones. In our project we also try to find out the potential fishing ground of a fish named sand eel.

Introduction

The North Sea is a semi-enclosed shallow shelf sea in northwestern Europe and one of the world's most productive seas. Between Great Britain, Norway, Denmark, Germany, the Netherlands, Belgium, and France lies a sea in the Atlantic Ocean known as the North Sea. The English Channel and the Norwegian Sea are its points of entry into the ocean, making it an epeiric (or "shelf") sea on the European continental shelf. It spans an area of 570,000 square kilometers and is more than 970 kilometers (600 miles) long and 580 kilometers (360 miles) broad (220,000 sq mi). It is surrounded by heavily populated countries with important industrial and agricultural activities, resulting in large quantities of nutrients and pollutants added to the North Sea. On the other hand, it has traditionally supported a significant fishery and hosted important North European maritime channels. In recent years, the sea has evolved into a rich supply of energy resources, including fossil fuels, wind, and early attempts at wave power. The shoreline is a well-liked location for relaxation and tourists in neighboring nations.

North sea has a huge variation of sea surface temperature & chlorophyll concentration. It may vary from month to month , season to season , year to year.

In our study area we are trying to find out the monthly sea surface temperature variation, chlorophyll concentration of North sea. we are also trying to find out the potential fishing ground of Sand ell.

Objectives

- 1) To find out the monthly SST (Sea Surface Temperature) variation of North Sea.
- 2) To find out the Chlorophyll concentration variation of North Sea.
- 3) To find out the potential fishing ground of Sand ell.

METHODS

Study area:

The North Sea is a sea of the Atlantic Ocean between Great Britain, Norway, Denmark, Germany, the Netherlands, Belgium and France. An epeiric (or "shelf") sea on the European continental shelf, it connects to the ocean through the English Channel in the south and the Norwegian Sea in the north. It is more than 970 kilometers (600 mi) long and 580 kilometers (360 mi) wide, covering 570,000 square kilometers (220,000 sq mi)

Coordinate of our study area (North sea) is

Min. Lat: 50° 59' 43.3" N (50.9954°)

Min. Long: 4° 26' 43.3" W (-4.4454°)

Max. Lat: 61° 1' 1.3" N (61.017°)

Max. Long: 12° 0' 21.4" E (12.0059°)

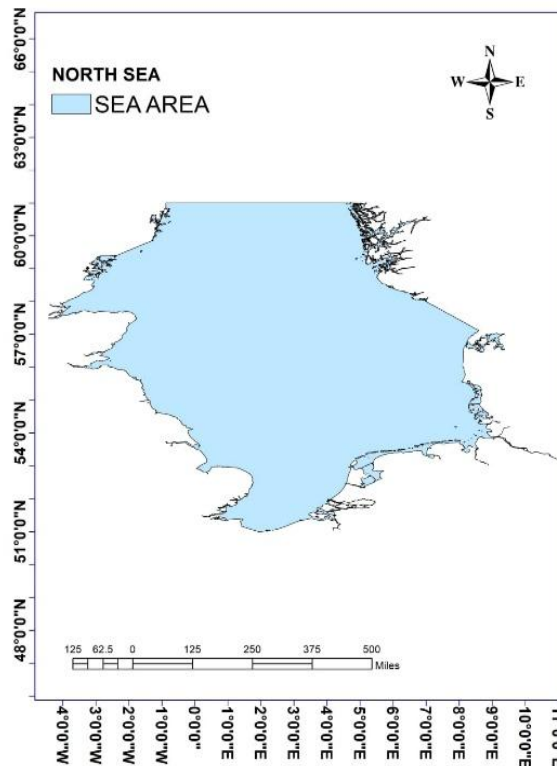


Figure: North Sea

Data collection:

We collect our data from some web site. They are Marineregions.org, ERDDAP, National Centers for Environmental Information & Ocean color web. The links are given below:

- 1) <https://www.marineregions.org>
- 2) <https://coastwatch.pfeg.noaa.gov/erddap/index.html>

- 3) <https://www.ncei.noaa.gov/products>
- 4) <https://oceancolor.gsfc.nasa.gov>

project activities

SST: Sea surface temperature (SST), or ocean surface temperature, is the water temperature close to the ocean's surface. Sea surface temperature may vary in different season. In warm season the sea surface temperature is higher than the winter season. In our project we use 2 month data of North sea. They are April 2021 & May 2021. Then we find out the mean of this two month.

Data interpolation:

The method of data interpolation is given below:

- Open Arc gis
- Make iho
- Input data
- Spatial Analyst tool
- Interpolation
- Kriging
- Input point feature
- Select environment
- Processing extent, raster analysis
- Done
- Export map

Our graph are given below.

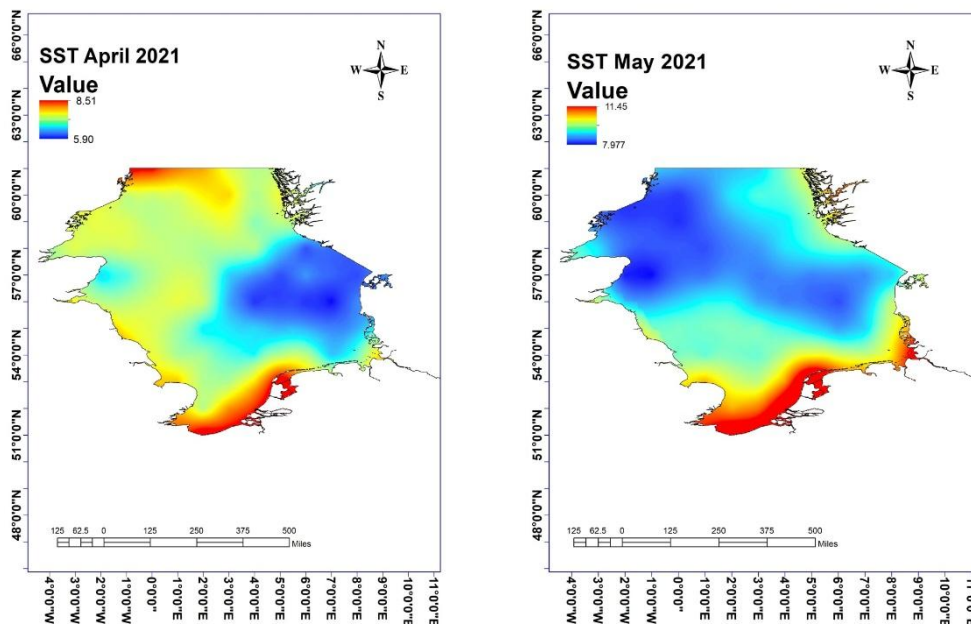


Figure: SST of April & May month 2021

Mean of this 2 month data is given below:

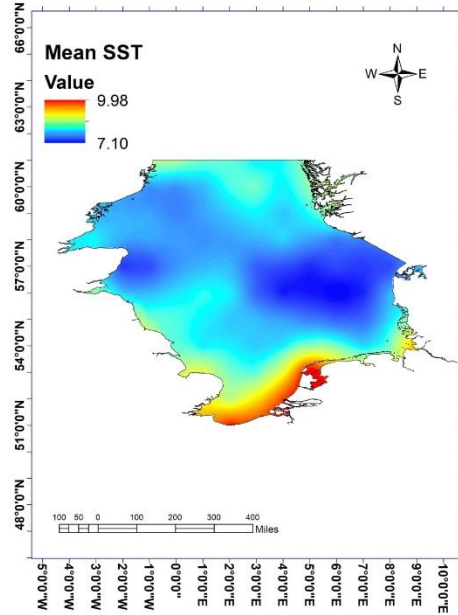


Figure: Mean SST of April & May 2021

We can classified mean SST to find which regions are warmer than other study point. Classification of mean SST of April & May are given below.

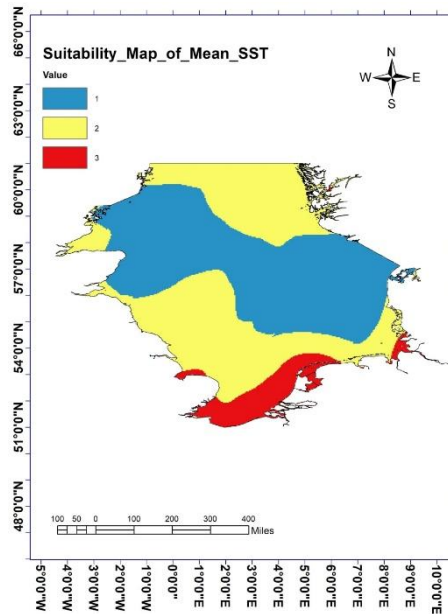


Figure: Reclassification of mean SST

Chlorophyll Concentration:

The concentration of chlorophyll is an indicator for the number of photosynthetic plankton, or phytoplankton, present in the ocean. Phytoplankton populations are influenced by climatic factors such as sea surface temperatures and winds. In the ocean, chlorophyll is required for measuring microscopic organisms as ocean is filled with this. In our project we use 2 months data of North Sea. They are April 2021 & May 2021. Then we find out the mean of this two month.

Data interpolation:

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Our graphs are given below:

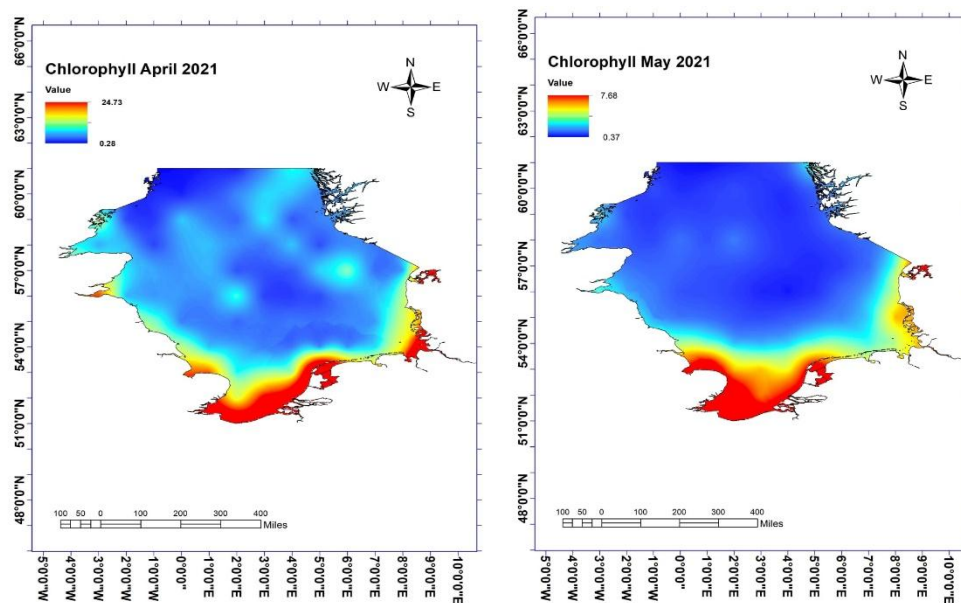


Figure: Chlorophyll of April & May month 2021

Mean of this 2months data is given below:

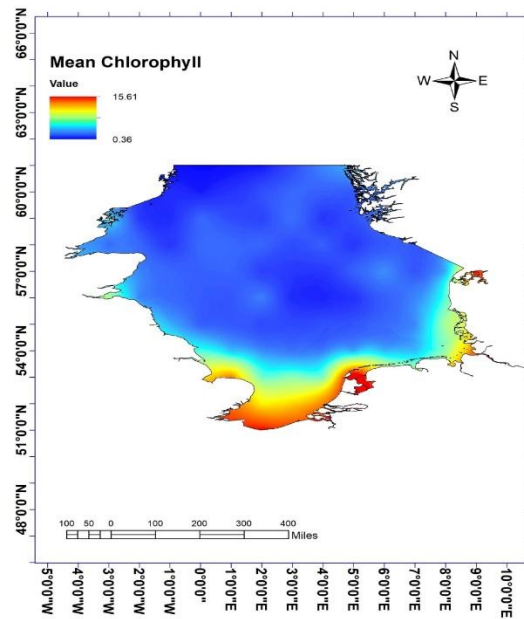


Figure: Mean SST of April & May 2021

Classification of mean SST of April & May are given below.

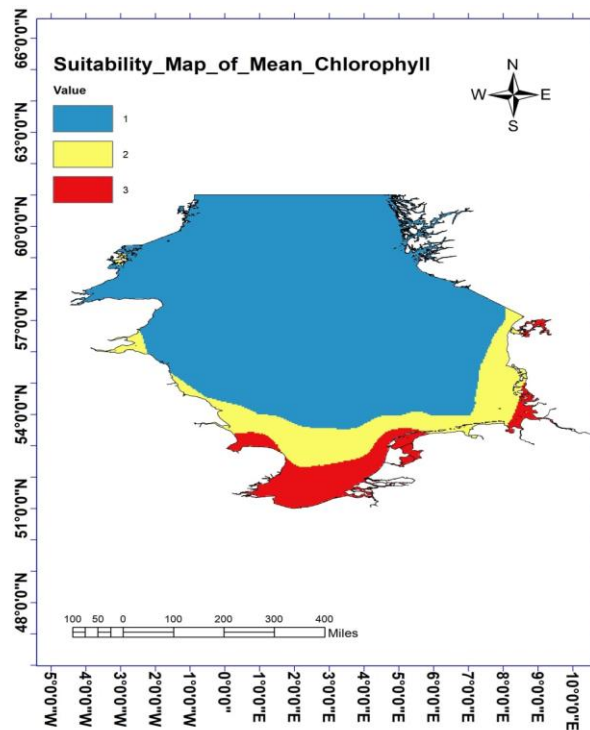


Figure: Reclassification of mean SST

Determination of potential fishing ground: with using georeferencing we find a map of potential fishing ground of sand ell. The map is given below:

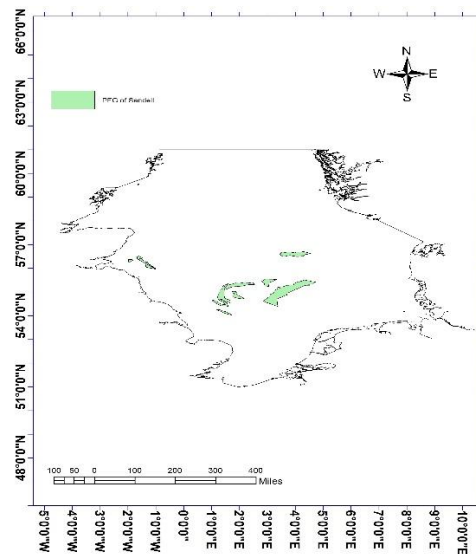


Figure: potential fishing ground of sand ell

Analysis

Sea Surface Temperature: In our study area (North Sea) we are trying to find out the variation of sea surface temperature. Sea surface temperature can vary from month to month, season to season, year to year, decades to decades. To learn about SST variation of North sea we collect data in created interpolation of data.

From this graph of SST April 2021, we see that maximum temperature of North Sea UK (North Atlantic region) & German part of North sea. Here we find temperature 8-8.51°C. But Danish part is colder than other region. The temperature is around 6°C. In mid ocean the Temperature is around 7°C.

In the graph of SST May 2021, we see that the maximum temperate region is German part of North Sea. The temperature is around 11°C in this region. But the UK part is colder part, the temperature is around 8°C. The mid region of sea has temperature around 10°C.

In mean SST graph we see that only German part is more warmer (10°C) than any other part of this sea. We see that the Danish part is relatively cold with temperature is around 7°C.

If we classified this sea in 3 part with SST variation we see that

- UK & Danish part is Cold region.
- Norway & English Channel part is warmer than UK & Danish part.
- German part is most warmer than any other part.

Chlorophyll: In our study are we also trying to fins out the chlorophyll concentration of North Sea. Chlorophyll concentration mainly depends on nutrient, river discharge, higher rainfall, salinity etc. In North

Sea, English Channel & Norwegian sea is the main source of River discharge. For this this part has relatively more chlorophyll concentration.

In this graph of chlorophyll concentration of April 2021, we see that, English channel, German part & Norwegian sea part are rich in chlorophyll concentration. This region has around 24 mg/L chlorophyll concentration but the UK part & North Atlantic Oceanic region has only 0.25-1 mg/L chlorophyll concentration.

From the graph of chlorophyll concentration of May in North sea. We can see the variation of chlorophyll concentration. This month we see that maximum concentration is 7.68 mg/L & minimum is 0.37 mg/L. In German & English channel has rich abundance of chlorophyll. This region has around 8mg/L concentration. In Norwegian part has also high abundances of chlorophyll concentration. But the Oceanic part is relatively poor in chlorophyll.

In mean chlorophyll concentration graph we see that German part is richest part & North Atlantic part is poor in abundance of Chlorophyll concentration.

If we classified this graph in 3 class. We see that

- German, Danish part & English channel is richest part in abundance of chlorophyll.
- Norwegian part has medium level of chlorophyll concentration
- UK & North Atlantic Ocean part has low chlorophyll concentration.

Potential Fishing Ground of Sand-ell: in our project we also trying to find out the habitat map of Sand ell fish. Using geo referencing we get the fishing ground of Sand ell. We find the PFG of sand ell near Germany, English Channel region. We find 6 fishing ground of Sand-ell.

Result

From our project we can reach in a discussion that ,

- German part of North sea is warmer than other region of North sea & it has a rich abundances of chlorophyll concentration.
- The UK & North Atlantic Ocean region has relatively low temperature & low chlorophyll concentration.
- The mid region of North sea has relatively medium level of SST& chlorophyll concentration.
- The potential fishing ground of Sand ell is near Germany & English channel.

Conclusion

In our project we find the monthly variation of sea surface temperature. We find a huge variation of sea surface temperature from a region to other & from month to month. We also find out the variation of chlorophyll concentration of North sea. That also varies from a location to other region. After all we can say that German region & English Channel region is rich in chlorophyll concentration & has high temperature relatively other region.